
Beyond Fitness Curves: Visualizing Population Dynamics in Neuroevolutionary Search

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Abstract

1 Neuroevolutionary algorithms maintain entire populations of candidate solutions
2 across generations, yet their dynamics are typically analyzed only through scalar
3 fitness metrics, which discard emergent geometric structure in the solution space.
4 We propose an analysis approach based on temporally aligned UMAP embeddings
5 and population velocity fields, with a post-hoc alignment procedure that reduces
6 arbitrary drift between successive projections, making temporal comparisons con-
7 sistent. Across two benchmarks with qualitatively distinct search regimes—Make
8 Moons ($d_w = 48$) and CIFAR-10 ($d_w \approx 197,000$)—we observe coherent popula-
9 tion flows associated with convergence basins, spatial localization of high-fitness
10 individuals, and task-dependent multimodal structure invisible in scalar metrics.
11 We show that these structures are largely obscured in unaligned projections, while
12 embedding-space velocity fields expose interpretable dynamic regimes through
13 spatial divergence patterns, suggesting that population geometry provides a com-
14plementary lens for understanding exploration, convergence, and multimodality in
15neuroevolution.

16 1 Introduction

17 Neuroevolutionary algorithms [23, 24] optimize neural network parameters through population-
18 based search, maintaining a diverse set of candidate solutions across generations. Despite their
19 widespread use, our understanding of their internal dynamics remains largely limited to scalar
20 summaries—mean fitness curves and best-individual trajectories—that collapse the population’s full
21 geometry into a single number per generation.

22 This is a significant blind spot. A population of neural networks is a cloud of points in a high-
23 dimensional weight space, and its evolution—convergence basins, exploratory spread, multimodal
24 structure—constitutes rich information about how search is proceeding. Scalar fitness curves may
25 not distinguish between a population converging toward a single attractor and one that is simultane-
26ously exploring multiple disconnected fitness peaks.

27 **Our approach.** We propose analyzing neuroevolution through temporally aligned low-
28 dimensional embeddings of evolving weight populations. At each generation, we project the popu-
29 lation into 2D using UMAP [15, 14] with a post-hoc alignment step that reduces arbitrary between-
30 generation drift, making successive projections directly comparable. From these aligned embed-
31 dings we derive *population velocity fields*—vector fields that summarize the average displacement
32 of individuals in embedding space from one generation to the next.

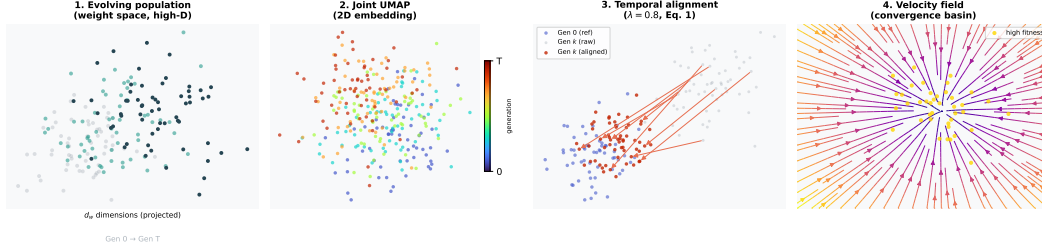


Figure 1: **Analysis pipeline.** At each generation, the neuroevolutionary population (a cloud of weight vectors) is projected jointly into 2D via UMAP. A post-hoc alignment step (Eq. 1, $\lambda = 0.8$) reduces arbitrary between-generation drift, making successive projections directly comparable. From the aligned per-generation embeddings we derive a *population velocity field*: for each cell of a 22×22 grid, the mean individual displacement from generation k to $k+1$ is computed and rendered as a streamline. The resulting visualization reveals convergence basins, multimodal structure, and fitness localization that are invisible in scalar fitness curves. This pipeline adapts the projection-trajectory vector field methodology of [1] from single-network hidden activations to evolving weight populations.

Contributions. Using this methodology, we identify three population-level geometric phenomena that have no analogue in single-network trajectory analysis, where there is no selection pressure, no competition between individuals, and no emergent multimodality across a population:

- (i) Coherent flow structure.** Population trajectories exhibit organized directional flow rather than isotropic stochastic spread, with identifiable convergence basins visible across multiple independent random seeds. This collective structure is driven by selection and cannot be observed by tracking a single network.
- (ii) Selection-driven fitness localization.** High-fitness individuals concentrate in compact, spatially coherent regions of the embedding. This spatial clustering emerges from selection pressure acting across the population—a phenomenon absent when visualizing a single network’s internal states.
- (iii) Distinct flow regimes across benchmarks.** The two benchmarks exhibit qualitatively different population geometries: Make Moons ($d_w = 48$, hidden=16) shows compact single-basin flow, whereas CIFAR-10 ($d_w \approx 197,000$, hidden=64) shows persistent multi-attractor structure. We report this as an empirical observation; we do not attribute it to any single factor (task complexity, dimensionality, mutation scale, or architecture) given that all differ between benchmarks (Section 6). This geometric difference is invisible in scalar fitness curves.

We additionally validate that these structures are not artifacts of the UMAP projection by (a) comparing aligned vs. unaligned projections and alternative dimensionality reduction baselines, (b) ablating the alignment strength, and (c) verifying that known synthetic cluster structure is faithfully recovered in UMAP embeddings.

2 Background

2.1 Neuroevolution

Neuroevolutionary algorithms represent neural network weights as genomes and apply evolutionary operators—mutation, selection, occasionally crossover—to optimize a fitness function evaluated on the network’s behavior [5, 23]. A central design choice is maintaining a *population* of individuals, enabling implicit parallelism and resistance to local optima. We focus on mutation-only evolution with elite selection, a simple but effective baseline used in several recent large-scale studies [24, 22].

62 2.2 Dimensionality Reduction for High-Dimensional Data

63 UMAP [15] is a nonlinear dimensionality reduction method that preserves local and global structure
 64 through a graph-based approximation to manifold learning. Unlike PCA, it can represent non-linear
 65 manifold structure; unlike t-SNE [25], it scales better to large datasets and preserves more global
 66 geometry. Both have been used to visualize neural network weight spaces [13, 7] and hidden ac-
 67 tivations [19, 1, 8], but not for tracking *populations* of independent networks across evolutionary
 68 generations.

69 2.3 Temporal Alignment of Projections

70 A fundamental challenge in applying dimensionality reduction to sequential data is that each in-
 71 dependent projection may differ by an arbitrary rotation, reflection, or translation, making direct
 72 comparison across time steps unreliable [14]. We address this with a *reference-anchored interpola-*
 73 *tion*: all generations are projected jointly into a single UMAP embedding (ensuring a shared global
 74 coordinate system), then each generation’s coordinates are interpolated toward the first generation’s
 75 embedding:

$$\tilde{\mathbf{Z}}_k = (1 - \lambda) \mathbf{Z}_k + \lambda \mathbf{Z}_{\text{ref}}, \quad (1)$$

76 where $\mathbf{Z}_k \in \mathbb{R}^{n \times 2}$ is the embedding of generation k , $\mathbf{Z}_{\text{ref}}$ is the reference (gen 0), and $\lambda \in [0, 1]$
 77 controls regularization strength. At $\lambda = 0$ the raw joint embedding is used; at $\lambda = 1$ all generations
 78 collapse to the reference. We select $\lambda = 0.8$ based on an ablation study (Section 4.3).

79 We use joint embedding with post-hoc interpolation rather than Aligned UMAP [14] because
 80 Aligned UMAP requires consecutive-generation correspondence during graph construction, whereas
 81 our approach preserves a single global coordinate system and allows arbitrary subsets of generations
 82 to be analyzed or re-aligned without recomputing the full embedding.

83 3 Methodology

84 3.1 Neuroevolution Setup

85 We evolve one-hidden-layer MLPs using mutation-only evolution with elite selection. At each gen-
 86 eration, the top $\alpha = 20\%$ of individuals (by fitness) are retained; the remainder are replaced by
 87 mutated copies of the elite, where mutations are sampled $\delta w \sim \mathcal{N}(0, \sigma^2)$ independently per weight.
 88 We study two benchmarks (Table 1):

Table 1: Benchmark configurations. d_w = total weight dimensions; σ = mutation standard deviation.

Benchmark	n pop	Gens	Hidden	σ	d_w	PCA pre-reduction
Make Moons	50	80	16	0.11	48	None
CIFAR-10	20	40	64	0.05	$\approx 197\text{k}$	50 dims

89 **Make Moons** is a 2D binary classification task from `sklearn` [17] with noise $\sigma = 0.1$. Fitness
 90 is negative cross-entropy loss on the full dataset. **CIFAR-10** [11] is a 10-class image classification
 91 task. Due to the high weight dimensionality ($d_w \approx 197,000$), we apply PCA to 50 dimensions
 92 before computing UMAP embeddings, following standard practice for high-dimensional weight
 93 visualization [13].

94 We deliberately study small-scale settings (one-hidden-layer MLPs, populations of 20–50) to enable
 95 rigorous multi-seed analysis and synthetic validation at tractable compute cost; scaling to larger
 96 populations and deeper architectures is left to future work. Code and instructions to reproduce all
 97 experiments are available at <https://anonymous.4open.science/r/neurips-2026-9729>.

98 3.2 Temporally Aligned UMAP Embeddings

99 All individuals across all generations are concatenated into a single matrix and projected jointly by
 100 UMAP (`neighbors = 15`, `min_dist = 0.1`), ensuring a consistent global coordinate system. Generation
 101 labels are preserved for color-coding. The alignment of Equation 1 is then applied point-wise using
 102 the first generation as reference.

3.3 Population Velocity Fields

From the aligned per-generation embeddings, we construct a 2D velocity field summarizing population flow. The embedding space is partitioned into a 22×22 grid; for each cell, we compute the mean displacement vector of individuals that are present in both generation k and $k + 1$ and whose centroid falls in that cell. The resulting field is smoothed with a 3×3 box kernel before visualization.

3.4 Metrics

Embedding spread (σ_1, σ_2) is the standard deviation of aligned embedding coordinates along each axis, measuring how dispersed the population is in projection space. Larger spread indicates greater dispersion; whether this reflects multimodality or slow convergence is disambiguated by the velocity field visualization (Section 3.3).

Temporal coherence measures how consistently the arrangement of individuals is preserved across successive projections:

$$\text{TC} = \frac{1}{T-1} \sum_{k=1}^{T-1} \left| \text{Pearson}(\tilde{\mathbf{Z}}_k, \tilde{\mathbf{Z}}_{k+1}) \right|, \quad (2)$$

where Pearson correlation is taken over flattened embedding vectors. Absolute value is used because UMAP may produce arbitrary reflections between independently projected generations.

Convergence rate is the first generation at which mean fitness reaches 95% of its final improvement: $g^* = \min\{g : \bar{f}_g \geq \bar{f}_0 + 0.95(\bar{f}_T - \bar{f}_0)\}$.

4 Results

4.1 Exp 1: Multi-Seed Robustness

To verify that observed structures are not artifacts of a particular random initialization, we run both benchmarks with 5 independent seeds (42, 123, 7, 31, 99) and compare the resulting embeddings and metrics (Table 2).

Table 2: Multi-seed robustness (mean $\pm$ std, 5 seeds). Spread σ_1 is the standard deviation of the aligned UMAP embedding along the first axis; convergence rate is the generation at which 95% of fitness improvement is achieved. Welch’s t -tests (Make Moons vs. CIFAR-10): σ_1 : $t(5.0) = -5.84$, $p = 0.0021$, $d = 3.69$; convergence: $t(6.7) = 5.37$, $p = 0.0012$, $d = 3.40$.

Benchmark	Embedding spread σ_1	Final fitness	Convergence (gen)
Make Moons	1.97 ± 0.08	-0.171 ± 0.006	47 ± 6
CIFAR-10	2.62 ± 0.22	-2.490 ± 0.022	20 ± 10

Make Moons shows compact, consistent embedding spread ($\sigma_1 = 1.97 \pm 0.08$) across all 5 seeds, with very low variance confirming that the near-unimodal structure is a robust property of the search process. CIFAR-10 exhibits substantially larger spread ($\sigma_1 = 2.62 \pm 0.22$), consistent with a more dispersed, multimodal population geometry. The difference is statistically significant despite the small sample: Welch’s t -test yields $t(5.0) = -5.84$, $p = 0.0021$, Cohen’s $d = 3.69$ (large effect). Convergence is faster for CIFAR-10 (20 ± 10 generations vs. 47 ± 6 ; $t(6.7) = 5.37$, $p = 0.0012$, $d = 3.40$) but to a substantially lower fitness level (-2.49 vs. -0.17), suggesting rapid collapse into one of several local basins rather than sustained global exploration. The high convergence variance for CIFAR-10 (± 10 gens) further reflects seed-dependent dynamics in the multimodal landscape. The velocity fields (Figure 2) confirm these differences visually: Make Moons shows radial convergence toward a single compact region, while CIFAR-10 shows branching flows toward multiple coexisting sink regions across all seeds.

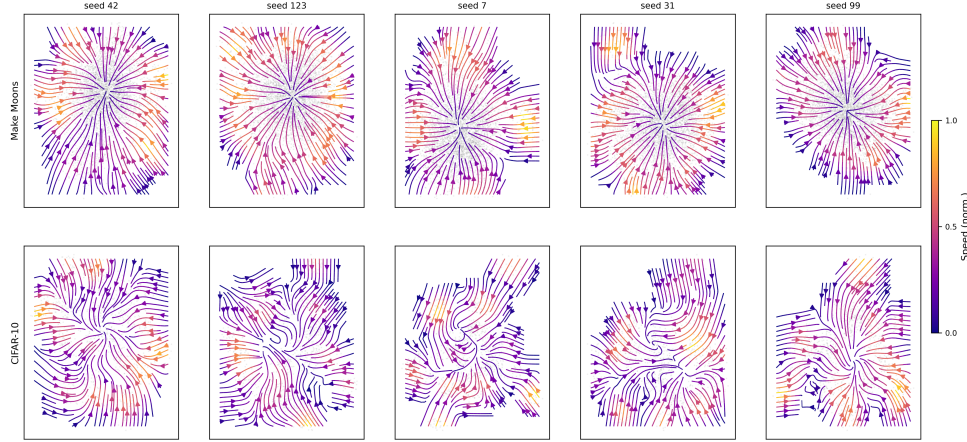


Figure 2: **Multi-seed reproducibility of population flow patterns.** 2 rows (Make Moons top, CIFAR-10 bottom) $\times$ 5 columns (seeds 42, 123, 7, 31, 99). Make Moons consistently shows radial convergence toward a single attractor ($\sigma_1 = 1.97 \pm 0.08$; $p = 0.0021$ vs. CIFAR-10, Welch’s t -test). CIFAR-10 reproduces 4–5 coexisting attractors per seed ($\sigma_1 = 2.62 \pm 0.22$); attractor locations vary across seeds, but the multi-sink topology is consistent. CIFAR-10 seed 31 (column 4) has an anomalously low scalar convergence rate (gen 3); see Table 5[†] for explanation. Together these support claims (i) and (iii).

136 4.2 Exp 2: Projection Baselines

137 We compare four projection methods on the Make Moons benchmark: PCA, t-SNE [25], UMAP
 138 unaligned, and UMAP aligned ($\lambda = 0.8$). For each method, we visualize generation coloring,
 139 fitness coloring, and the derived velocity field.

140 PCA recovers a two-moons-like structure in weight space (Figure 3, top row), reflecting input-space
 141 geometry that bleeds into early-layer weights; this is static dataset-induced structure, not search
 142 dynamics. PCA conflates generations without revealing temporal flow. t-SNE fragments the embed-
 143 ding into many small disconnected clusters (attractor count = 298 by DBSCAN), making velocity
 144 fields uninterpretable. UMAP unaligned shows more structure (attractor count = 64) but exhibits
 145 arbitrary between-generation drift that distorts velocity fields. UMAP aligned ($\lambda = 0.8$) produces
 146 the most compact and temporally coherent embedding (attractor count = 21, spread $\sigma_1 = 1.88$),
 147 with smooth velocity fields that clearly reveal convergence dynamics. (The spread here uses seed 42
 148 only; the cross-seed mean is 1.97 ± 0.08 , Table 2.)

149 4.3 Exp 3: Alignment Strength Ablation

150 We sweep $\lambda \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ on the Make Moons benchmark and measure temporal
 151 coherence and structural compactness. Results are shown in Figure 4 and Table 3.

Table 3: Exp 3: alignment ablation on Make Moons. Temporal coherence and embedding spread as a function of λ .

λ	Temporal coherence	Attractor count	Spread σ_1
0.0	−0.010	10	8.58
0.2	0.005	10	6.88
0.4	0.045	10	5.20
0.6	0.169	7	3.53
0.8	0.587	5	1.88
1.0	1.000	10	0.93

152 Temporal coherence increases monotonically with λ , reaching TC = 1.0 at $\lambda = 1.0$ (all embeddings
 153 collapse to the reference by construction). However, $\lambda = 1.0$ eliminates all temporal variation and

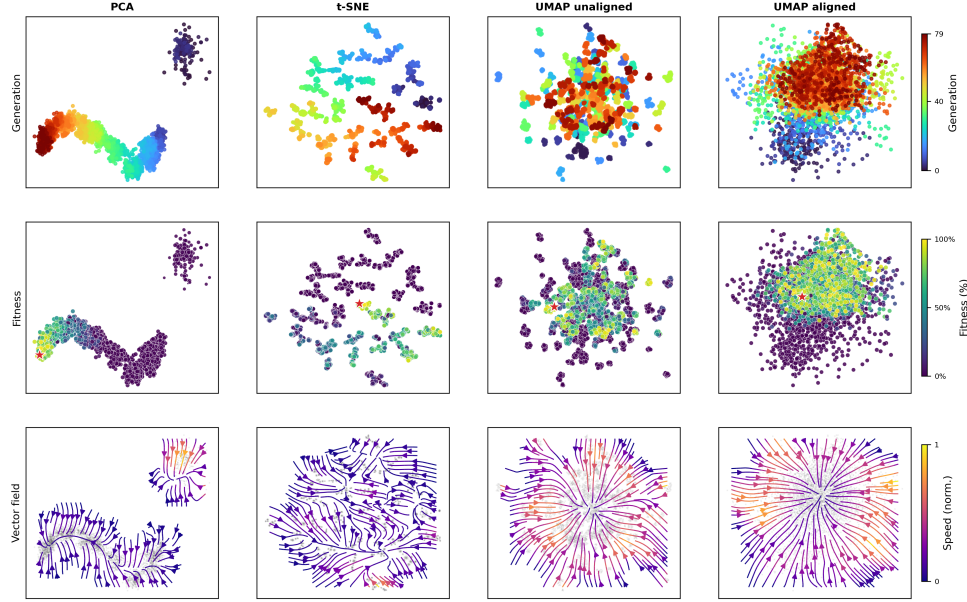


Figure 3: Exp 2: four projection methods on Make Moons (80 generations, 50 individuals). Columns: PCA, t-SNE, UMAP unaligned, UMAP aligned. Rows: generation coloring (top), fitness coloring (middle), velocity field (bottom). Only UMAP aligned ($\lambda = 0.8$) produces interpretable temporal structure and smooth velocity fields.

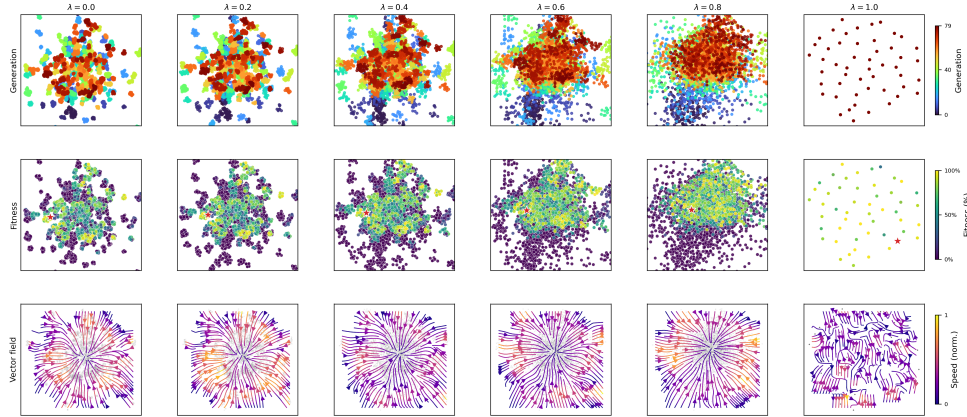


Figure 4: Exp 3: λ sweep ($\lambda \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, columns). Rows: generation coloring (top), fitness coloring (middle), velocity field (bottom). See Figure 6 for the TC vs. λ curve.

154 compresses the embedding ($\sigma_1 = 0.93$), destroying interpretable structure. The value $\lambda = 0.8$
155 provides the best trade-off: high coherence (TC = 0.587) with meaningful spread ($\sigma_1 = 1.88$), and
156 velocity fields that visually reveal distinct convergence dynamics (Figure 4); the full TC vs. λ curve
157 is shown in Appendix A.5 (Figure 6). Attractor count (Table 3) is provided as an ordinal indicator
158 only; as noted in Appendix A.3, it is sensitive to embedding density and should not be interpreted
159 as a precise geometric measurement.

160 4.4 Exp 4: Synthetic Validation

161 To verify that UMAP-detected structure reflects genuine data geometry rather than projection arti-
162 facts, we apply our pipeline to three synthetic datasets with known structure (Table 4): a single
163 Gaussian cluster, four well-separated Gaussian clusters, and a random walk (each generation inde-
164 pendently sampled from $\mathcal{N}(0, I)$, no temporal correspondence between points).

Table 4: Exp 4: synthetic validation. Temporal coherence computed on unaligned embeddings ($\lambda = 0$) to avoid inflation from the alignment procedure itself.

Dataset	Expected modes	Temporal coherence
Single Gaussian	1	0.29
Four Gaussians	4	0.91
Random walk	0	0.03

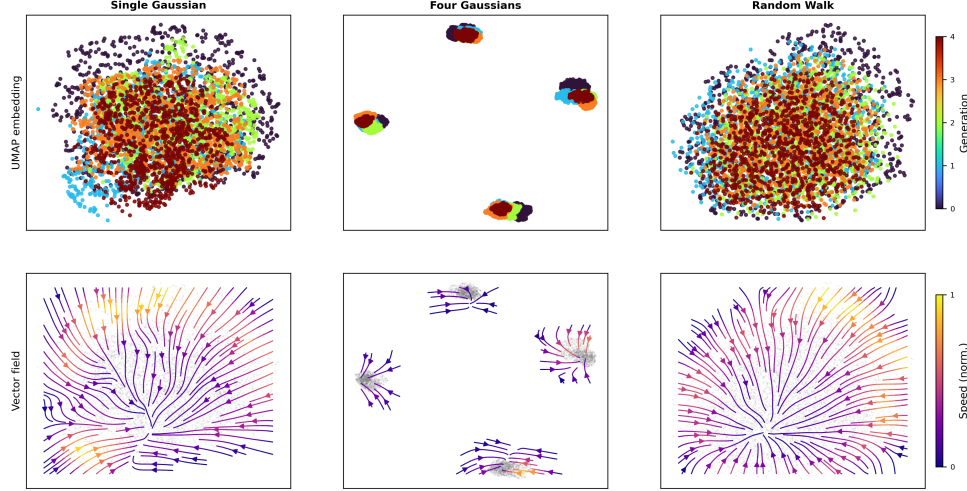


Figure 5: **Exp 4: synthetic validation of UMAP structure recovery.** Columns: Single Gaussian (left), Four Gaussians (center), random walk (right). Rows: UMAP embedding (top), velocity field (bottom). *Single Gaussian*: structureless data yields a diffuse, rotation-prone embedding (TC = 0.29). *Four Gaussians*: four well-separated clusters are faithfully recovered (TC = 0.91). *Random walk*: independently sampled point clouds produce no coherent structure (TC = 0.03), confirming absence of spurious UMAP artifacts.

165 The random walk yields near-zero temporal coherence (TC = 0.03), confirming that independent
 166 point clouds produce no spurious structure. The four-Gaussian dataset achieves high coherence
 167 (TC = 0.91), indicating that genuine multi-cluster structure is faithfully preserved across gener-
 168 ations. The single Gaussian shows intermediate coherence (TC = 0.29); this reflects UMAP’s
 169 tendency to produce arbitrary rotations between independent projections of structureless data—even
 170 when the underlying point cloud is temporally identical, successive projections rotate relative to each
 171 other, reducing the Pearson correlation in our TC metric. The four-Gaussian case avoids this because
 172 cluster identity provides anchor points that constrain rotation. Note that Table 4 reports TC at $\lambda = 0$
 173 by design, to isolate UMAP’s behaviour without alignment. With $\lambda = 0.8$ (used in Experiments
 174 1–3), the reference-anchored interpolation suppresses between-generation drift for all datasets in-
 175 cluding the Single Gaussian, confirming that alignment works as intended across structural regimes.
 176 The high TC values in Experiments 1–3 are therefore not inflated by alignment masking structureless
 177 data: when there is no structure, $\lambda = 0$ already gives near-zero TC (random walk: 0.03), and align-
 178 ment cannot manufacture coherence that is not present in the embeddings. Together, these results
 179 confirm that the structures observed in the neuroevolution experiments are not UMAP artifacts.

180 **5 Discussion**

181 Our results suggest that neuroevolutionary populations exhibit structured geometric dynamics in
 182 weight space that are not captured by scalar fitness metrics. The consistent attractor-like regions
 183 and coherent flow patterns across independent seeds indicate that the observed structure is a robust
 184 property of the search process, not an artifact of initialization or projection.

The task-dependent multimodality in claim (iii)—compact near-unimodal flow for Make Moons ($d_w = 48$, hidden=16) versus persistent multimodal structure for CIFAR-10 ($d_w \approx 197,000$, hidden=64)—is consistent with known properties of high-dimensional loss landscapes: larger networks have more saddle points and flat regions [4], which could support coexistence of multiple fitness basins simultaneously. However, we caution that the two benchmarks also differ in mutation rate ($\sigma = 0.11$ vs. 0.05), architecture depth, and task complexity; disentangling which factor drives the multimodality difference requires controlled experiments (Section 6).

The spatial localization of high-fitness individuals (claim ii) suggests that fitness landscape curvature is reflected in embedding geometry, which could enable early detection of promising regions before fitness curves have converged. This has potential implications for adaptive population management strategies.

Possible mechanisms for the multimodality difference. Several non-exclusive mechanisms could contribute to the observed difference in population geometry between benchmarks. First, permutation symmetry: in networks with hidden units, any permutation of hidden neurons produces a functionally equivalent network at a different weight vector, creating an exponential number of symmetric optima that grow with network size. Second, at high dimensionality, the probability that random mutations escape a local basin decreases—populations tend to specialize in distinct regions rather than spanning the full landscape. Third, mutation scaling: we use a fixed $\sigma = 0.05$ for CIFAR-10 vs. $\sigma = 0.11$ for Make Moons; smaller perturbations relative to weight magnitude may reduce mixing across basins. Disentangling these mechanisms is an important direction for future work.

Practical implications. The ability to observe convergence basins and multimodal structure in real time suggests several concrete applications. Practitioners could use velocity fields to detect premature convergence early, triggering diversity-restoring interventions before fitness plateaus. The spatial localization of high-fitness individuals could guide offspring allocation in non-uniform mutation strategies, concentrating exploration around identified attractor regions. More ambitiously, if embedding geometry at generation k predicts eventual convergence targets, it could enable early stopping or adaptive population sizing.

6 Limitations

UMAP stochasticity. UMAP embeddings depend on random initialization and neighbor graph construction. While our temporal alignment reduces between-generation drift, independent runs with different random seeds may produce qualitatively different embeddings for the same population. We mitigate this by using a fixed random seed for all projections and reporting results across 5 evolutionary seeds.

Embedding spread as a proxy for multimodality. We use embedding spread (σ_1) as the primary quantitative indicator of population structure. While spread correlates with multimodality visually, it is a coarse proxy: a dispersed embedding could reflect genuine multi-basin structure or simply a slower-converging population. The velocity field visualizations provide the primary evidence for claim (iii); spread serves as a reproducible scalar summary.

Scale of benchmarks. Our experiments use small populations and networks to keep compute tractable. It is unclear whether the observed phenomena scale to the population sizes and architectures used in state-of-the-art neuroevolution.

Mutation scaling confound. The two benchmarks use different mutation rates ($\sigma = 0.11$ for Make Moons, $\sigma = 0.05$ for CIFAR-10), chosen empirically to produce comparable per-generation fitness progress on each task. This confounds the dimensionality contrast in claim (iii): a smaller σ relative to weight magnitude may itself reduce cross-basin mixing, independently of dimensionality. A controlled experiment varying σ within a single benchmark while holding architecture fixed would help disentangle these effects.

Methodological scope. The vector-field-on-aligned-projections methodology is established in the context of single-network hidden activation visualization [1]; our contribution is in applying it to evolving weight *populations* and characterizing the distinct population-level phenomena that emerge, not in the projection or vector field machinery itself.

236 **Causal interpretation.** We observe *correlations* between embedding structure and fitness dynamics.
237 We do not claim that the geometric structure *causes* or *predicts* evolutionary outcomes; only that it
238 is consistently present and visually informative.

239 7 Related Work

240 **Neuroevolution.** Modern neuroevolution methods [24, 22, 21] have achieved strong performance
241 on RL benchmarks but remain poorly understood mechanistically. Our work is complementary
242 to fitness landscape analysis [10] and quality-diversity methods [16] that also seek to understand
243 population structure.

244 **Vector field visualization of projection trajectories.** Cantareira et al. [1] introduced the method-
245 ology of constructing vector fields from sequentially aligned projections to visualize hidden layer
246 activity in deep networks across training epochs and layers. Their approach combines a generic
247 projection alignment model [2] with Vector Field K -Means [6] for trajectory aggregation; Rauber
248 et al. [20, 19] established related trajectory-based visualization of hidden activations using dynamic
249 dimensionality reduction. Our work applies this family of techniques to a qualitatively different
250 regime: instead of tracking a single network’s internal states, we track *populations* of independent
251 networks across evolutionary generations. This shift introduces phenomena absent in single-network
252 visualization: selection pressure creates fitness-correlated spatial structure, attractor competition
253 emerges from parallel independent trajectories, and task-dependent search dynamics manifest at the
254 population level rather than within individual activation patterns.

255 **Weight space visualization.** Li et al. [13] visualize loss landscapes via filter-normalized linear inter-
256 polation; Goodfellow et al. [7] study loss surfaces along linear paths. These approaches characterize
257 a single network’s loss surface; our work characterizes the collective trajectory of a population. For
258 a survey of visual analytics in deep learning, see Hohman et al. [9].

259 **Dimensionality reduction for neural networks.** UMAP and t-SNE have been applied to visualize
260 neuron activations [3] and model families [18]. Our contribution is to apply temporally aligned
261 projection to evolving weight *populations*, revealing population-level geometric phenomena specific
262 to the evolutionary regime.

263 **Behavioral diversity in evolution.** MAP-Elites [16] and novelty search [12] explicitly maintain
264 behavioral diversity; our approach provides post-hoc geometric analysis of diversity in weight space
265 without requiring behavioral descriptors.

266 8 Conclusion

267 We have presented an analysis methodology for neuroevolutionary population dynamics based on
268 temporally aligned UMAP embeddings and population velocity fields. Applied across benchmarks
269 with qualitatively different search regimes, the method consistently reveals geometric structure—
270 coherent flow, fitness localization, and task-dependent multimodality—that is invisible in scalar
271 fitness metrics. Ablation and synthetic validation experiments confirm that the observed structure
272 reflects genuine evolutionary dynamics rather than projection artifacts. We release code and interac-
273 tive visualizations to facilitate adoption by the neuroevolution community (<https://anonymous.4open.science/r/neurips-2026-9729>).
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A Additional Experimental Details

A.1 Network Architecture and Training

All networks are one-hidden-layer MLPs with sigmoid activations. Make Moons: input $2 \rightarrow 16 \rightarrow 1$ (sigmoid output, binary cross-entropy). CIFAR-10: input $3072 \rightarrow 64 \rightarrow 10$ (softmax output, negative log-likelihood). Elite fraction = 20%, minimum elite = 2 individuals.

A.2 Compute Resources

All experiments run on a single CPU (Apple M-series, 16GB RAM). Approximate wall-clock times: Make Moons (5 seeds) $\approx$ 1 min; CIFAR-10 (5 seeds) $\approx$ 20 min; Exp 2–3 $\approx$ 1–2 min each; Exp 4 $\approx$ 1 min. No GPU required. Total reproduction time $\lesssim$ 30 min.

A.3 Full Multi-Seed Results

Table 5 reports all per-seed metrics including attractor count, estimated by DBSCAN with adaptive ε (90th percentile of k -NN distances, $k = 5$). Attractor counts are sensitive to embedding density and should be interpreted as rough ordinal indicators rather than precise geometric measurements; we use embedding spread as the primary metric in the main text.

Table 5: Per-seed metrics for all benchmarks. Attractor count uses DBSCAN with adaptive ε ; values vary with embedding density and are provided for completeness only. †CIFAR-10 seed 31 reports convergence at generation 3: the fitness trajectory began close to its final value under this initialization, so 95% of the negligible improvement span ($\bar{f}_0 \approx \bar{f}_T$) was reached trivially early. This is a numerical artifact of the convergence-rate definition when the fitness is nearly flat, not an unusually fast run; it is the primary driver of the high convergence variance (± 10) reported in Table 2.

Benchmark	Seed	Spread σ_1	Fitness	Conv. (gen)	Attractors
Make Moons	42	1.88	−0.169	57	19
	123	1.92	−0.171	41	13
	7	1.92	−0.163	47	13
	31	2.07	−0.175	45	13
	99	2.03	−0.179	45	18
CIFAR-10	42	2.45	−2.464	19	5
	123	2.31	−2.497	26	4
	7	2.79	−2.497	26	5
	31 [†]	2.64	−2.519	3	5
	99	2.89	−2.472	24	4

352 A.4 Hyperparameter Table

Table 6: Full hyperparameter table for all benchmarks.

Benchmark	Pop	Gens	Hidden	σ	λ	PCA dims
Make Moons	50	80	16	0.11	0.8	—
CIFAR-10	20	40	64	0.05	0.8	50

353 A.5 Temporal Coherence vs. Alignment Strength

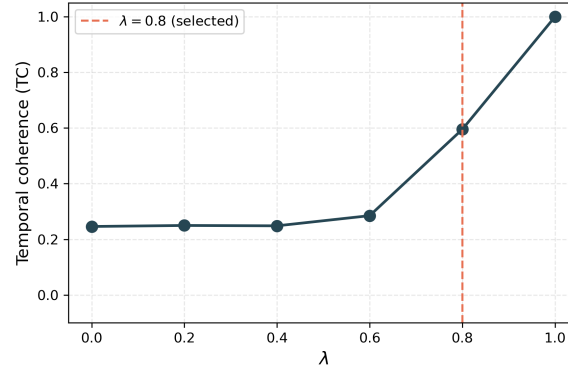


Figure 6: Temporal coherence as a function of alignment strength λ . Coherence increases monotonically; $\lambda = 0.8$ (dashed line) provides the best trade-off between coherence (TC = 0.587) and embedding diversity ($\sigma_1 = 1.88$). At $\lambda = 1.0$, all generations collapse to the reference (TC = 1.0) but temporal structure is destroyed.

354 A.6 Interactive Visualization Tool

355 We developed a browser-based interactive tool built with Streamlit that exposes all visualization
 356 components described in this paper—aligned UMAP projections and population velocity fields—
 357 through a parameter-driven interface. Users can configure population size, number of generations,
 358 hidden dimensionality, mutation rate, alignment strength λ , and toggle between static (Matplotlib)
 359 and interactive (Plotly) rendering modes in real time.

360 Figures 7–9 show both rendering modes on the Make Moons benchmark with $\lambda = 0.8$ and 80
 361 generations.

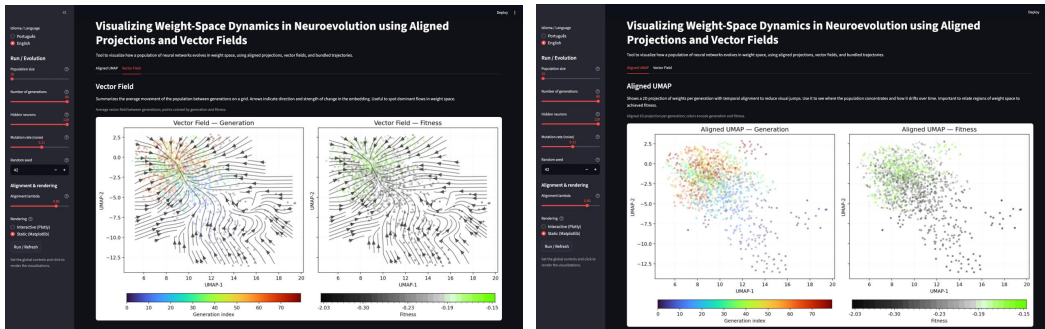


Figure 7: Static Matplotlib mode. **Left:** Vector Field tab—population flow colored by generation (left panel) and fitness (right panel). **Right:** Aligned UMAP tab—per-generation 2D embeddings with generation and fitness colorbars. The sidebar exposes all run parameters for rapid hypothesis testing.

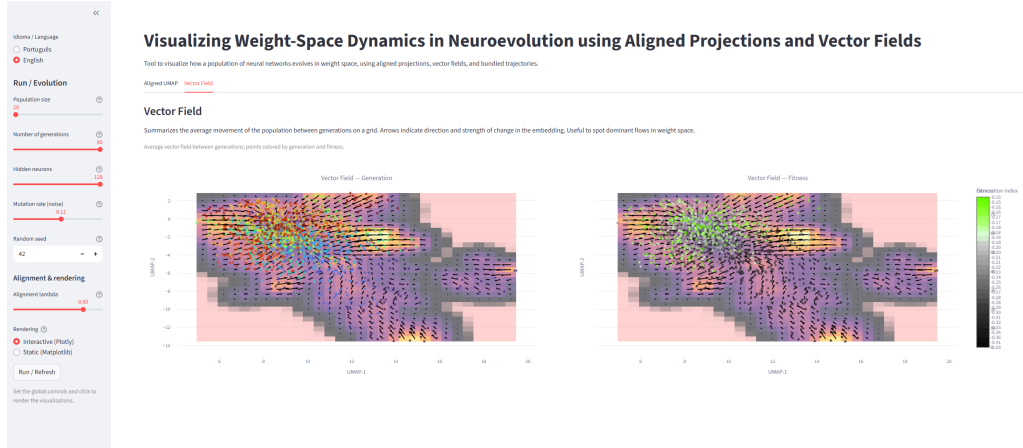


Figure 8: Interactive tool (Plotly mode) showing the Vector Field tab on Make Moons, $\lambda = 0.8$, 80 generations. Quiver arrows indicate population flow direction and magnitude; background density heatmap shows where individuals concentrate. Left: colored by generation index. Right: colored by fitness. The sidebar exposes all run parameters for rapid hypothesis testing.

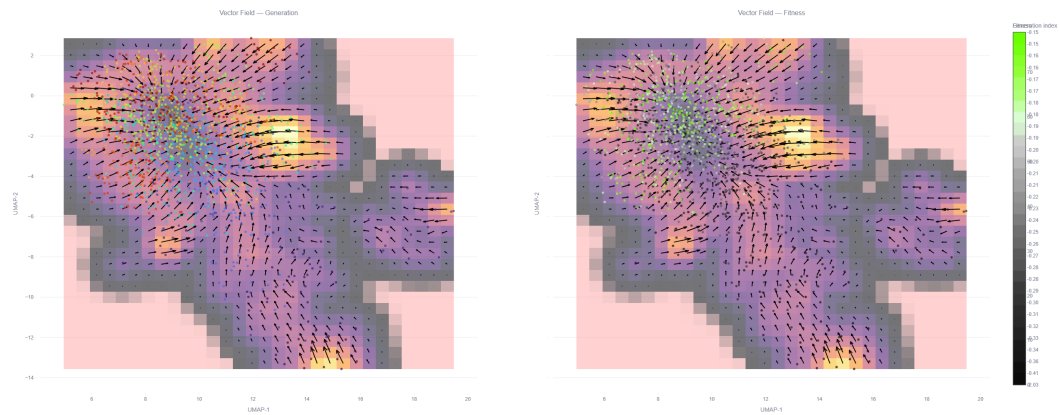


Figure 9: Detail view of the interactive Plotly quiver vector field. The density background and directional arrows together reveal the dominant convergence basin and the multimodal structure of the Make Moons fitness landscape. Points are individually hoverable; the plot supports zoom and pan.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state three observational claims (coherent flow, fitness localization, distinct flow regimes across benchmarks) that are directly supported by the experiments in Section 4. All claims are framed as empirical observations with appropriate hedging (“we observe,” “suggests,” “consistent with”), not as theoretical guarantees. The scope is explicitly limited to the two benchmarks described.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 (Limitations) discusses six explicit limitations: UMAP stochasticity and sensitivity to initialization; the use of embedding spread as a coarse proxy for multimodality; the small scale of benchmarks and populations; a mutation-scaling confound between the two benchmarks; the methodological scope (established methodology applied to a new domain); and the observational (non-causal) nature of the findings.

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Justification: The paper makes no theoretical claims and presents no proofs. All contributions are empirical observations.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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Justification: Section 3 and Appendix A describe all hyperparameters, network architectures, UMAP settings ($k = 15$, $\text{min_dist} = 0.1$), alignment procedure (Equation 1), and metric definitions. The full hyperparameter table is provided in Appendix A.3. Code is released with the submission.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code is provided as anonymized supplemental material. All benchmarks use publicly available datasets: Make Moons (scikit-learn), CIFAR-10 (Toronto/Krizhevsky 2009). A `pixi.toml` environment file and README are included for one-command environment setup.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3.1 and Table 1 specify population sizes, generations, hidden dimensions, mutation rates, and PCA pre-reduction. There is no gradient-based training; the optimizer is mutation-only evolution with elite selection ($\alpha = 20\%$). Appendix A.3 provides the full hyperparameter table for both benchmarks.

412 **7. Experiment statistical significance**

413 Question: Does the paper report error bars suitably and correctly defined or other appropri-
414 ate information about the statistical significance of the experiments?

415 Answer: [Yes]

416 Justification: Table 2 (Exp 1) reports mean $\pm$ standard deviation across 5 independent
417 random seeds. Welch’s t -tests between Make Moons and CIFAR-10 are reported in the Ta-
418 ble 2 caption and body text: embedding spread σ_1 : $t(5.0) = -5.84$, $p = 0.0021$, Cohen’s
419 $d = 3.69$; convergence rate: $t(6.7) = 5.37$, $p = 0.0012$, $d = 3.40$. The small sample size
420 ($n = 5$) is acknowledged as a limitation in Section 6.

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424 the experiments?

425 Answer: [Yes]

426 Justification: Appendix A.2 reports that all experiments run on a single CPU (Apple M-
427 series, 16 GB RAM) with approximate wall-clock times per experiment. No GPU is re-
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